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Engineer to Excel

Computer Vision - Introduction, Key Features, Pipeline, Levels, Importance

ITA0522 - COMPUTER VISION

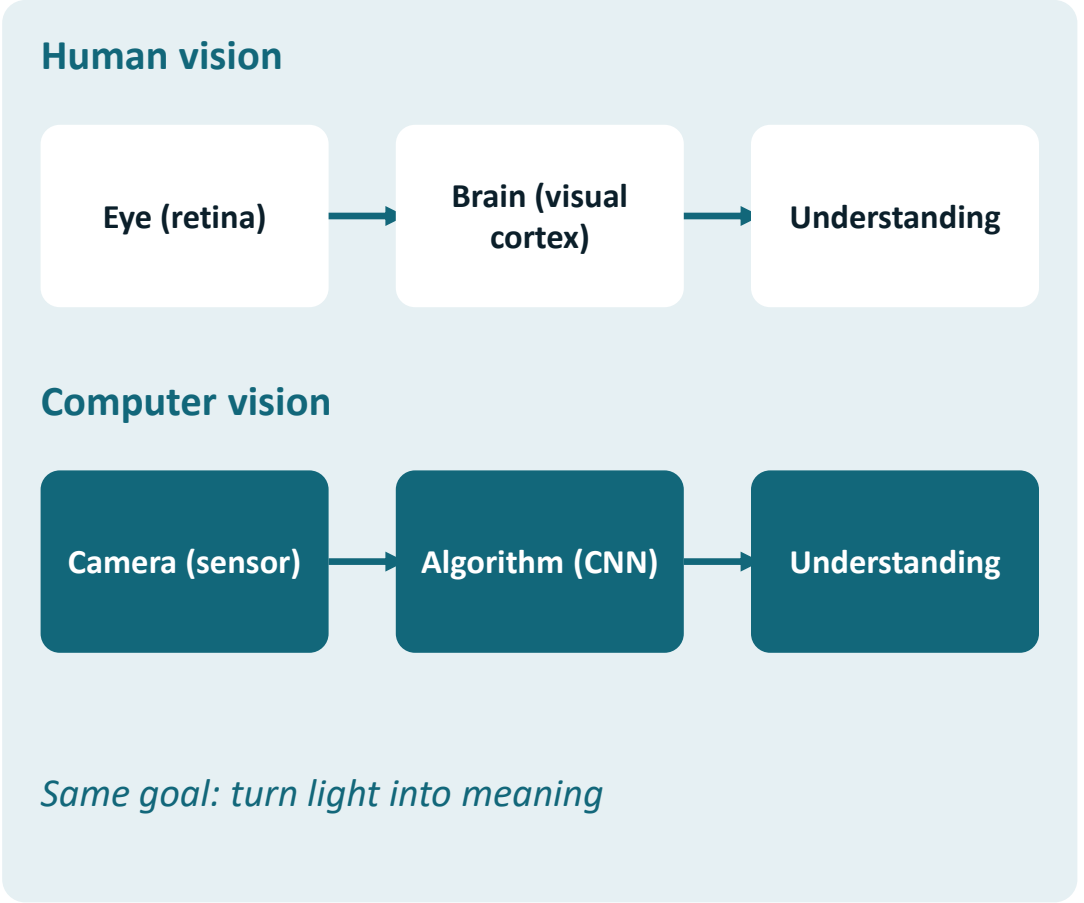
CO1 AT3 - TECHNICAL PRESENTATION

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Introduction to Computer Vision

- Computer Vision is a branch of Artificial Intelligence (AI) that enables computers to capture, process, and understand images and videos.
- It helps machines identify objects, recognize patterns, and interpret visual information from the real world.
- Computer Vision is widely used in applications such as facial recognition, medical imaging, self-driving cars, and industrial inspection.
- Its main objective is to enable computers to understand visual data and make intelligent decisions based on it.



Problem Context: Factory Defect Inspection

- **Problem:** Manual inspection of manufactured products is time-consuming and may lead to human errors.
- **Image Acquisition:** Cameras capture images of products moving along the production line.
- **Image Processing:** Computer Vision processes the images to improve quality and identify important features.
- **Defect Detection:** The system identifies defects such as cracks, scratches, dents, and incorrect dimensions.
- **Solution and Benefits:** Defective products are automatically identified and separated, improving quality, accuracy, and production efficiency.

Manual inspection

- Subjective judgement
- Fatigue-prone
- Hard to scale
- No data trail

Computer vision inspection

- Objective and repeatable
- Runs around the clock
- Scales across lines
- Logs every part

Key Features of Computer Vision

1

Image Classification

Assigns one label to a whole image

e.g. part is OK or defective

2

Object Detection

Finds objects and draws bounding boxes

e.g. locate each part on the belt

3

Segmentation

Labels every pixel to outline exact shapes

e.g. trace the crack boundary

4

Recognition & OCR

Identifies faces, text and codes

e.g. read serial numbers

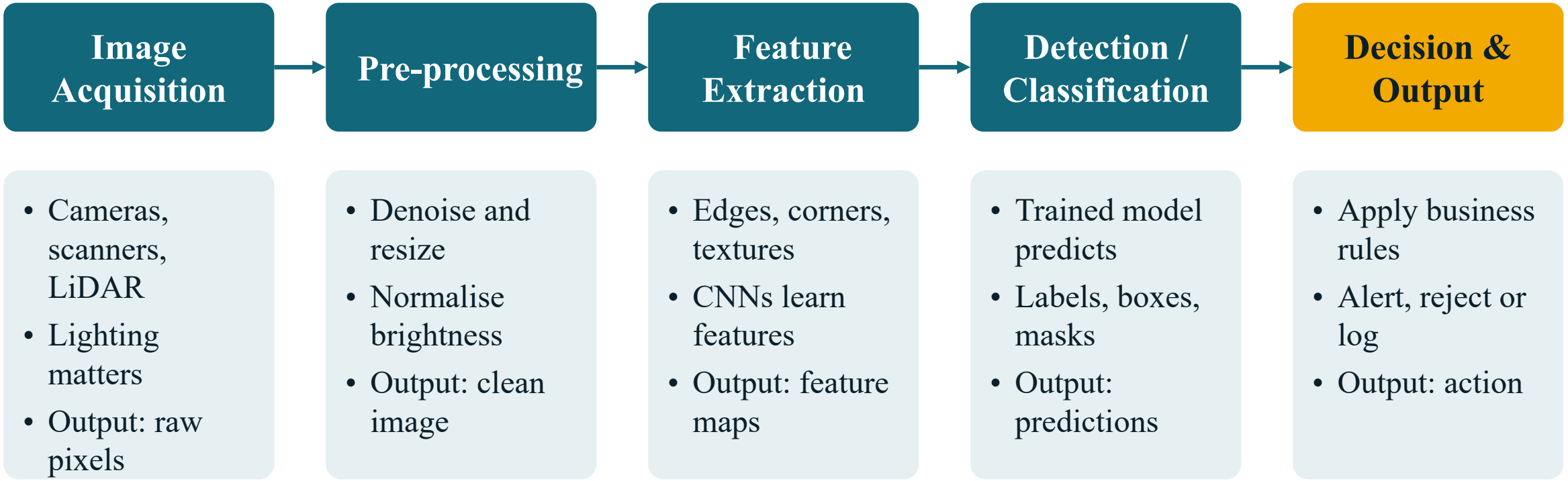
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Tracking & Motion

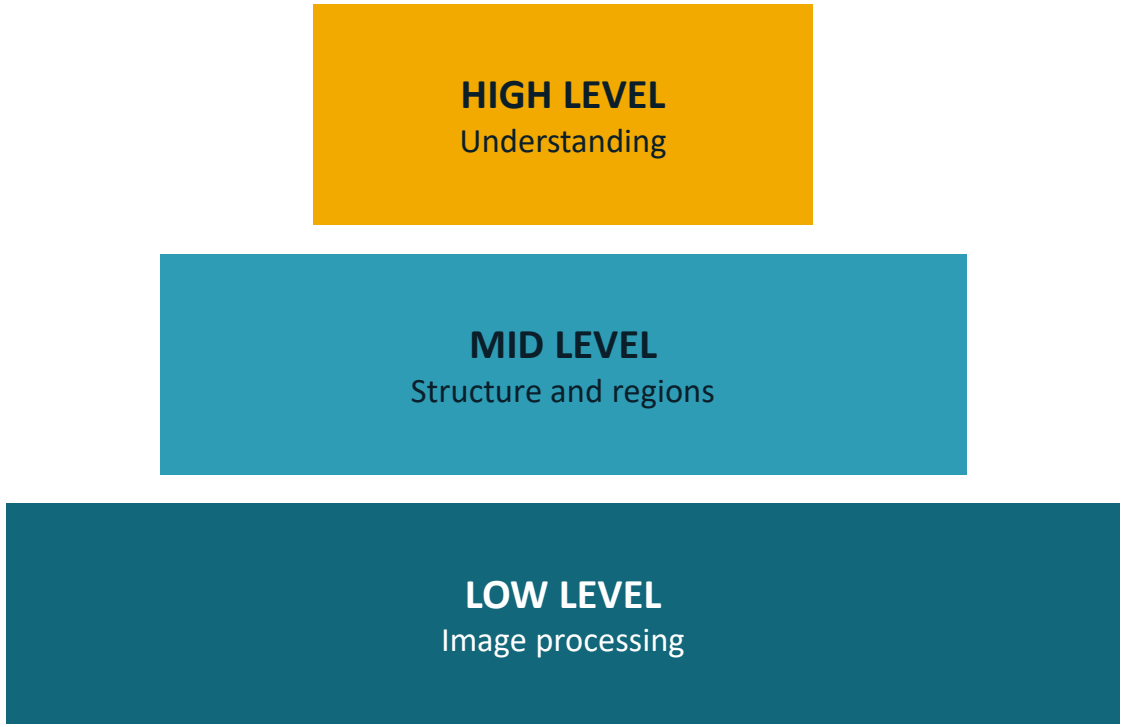
Follows objects across video frames

e.g. follow parts along the conveyor

The Computer Vision Pipeline



Levels of Computer Vision

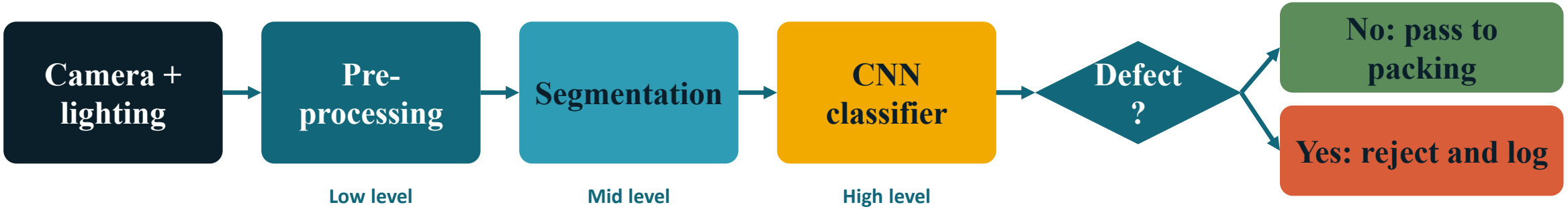


- **Low-Level Vision:** Processes raw images to improve quality by removing noise, adjusting brightness, and enhancing contrast.
- **Mid-Level Vision:** Extracts important features such as edges, shapes, textures, and object boundaries from images.
- **High-Level Vision:** Identifies objects, classifies images, and understands the meaning of visual scenes.
- **Application in Defect Inspection:** Low-level vision enhances product images, mid-level vision extracts defect features, and high-level vision classifies products as defective or non-defective.
- **Overall Importance**: These levels work together to improve image analysis, automate inspection, and increase manufacturing quality.

Levels Compared Side by Side

	Low level	Mid level	High level
Aim	Improve and prepare the image	Extract structure and regions	Interpret meaning and act
Data flow	Image to image	Image to features	Features to decisions
Techniques	Filtering, histogram equalisation, edge detection	Segmentation, contours, SIFT, optical flow	CNNs, YOLO, Vision Transformers
Example task	Remove noise, fix brightness	Separate part from background	Classify part as OK or defective
In the factory	Cut camera glare	Outline the crack region	Trigger the reject mechanism

Applying the Levels: Inspection System



Low level: removes glare and noise so defects stand out

Mid level: isolates the part and candidate defect regions

High level: the CNN labels each part as OK or defective in real time

Action: a diverter rejects faulty parts; every result is logged to a dashboard

Improvement: flagged images are fed back to retrain the model

Importance and Real-World Applications

- Automates product inspection and reduces the need for manual checking in factories.
- Improves accuracy by identifying defects such as cracks, scratches, dents, and surface irregularities.
- Saves time by inspecting large numbers of products quickly on production lines.
- Reduces manufacturing costs by minimizing human errors, material waste, and defective products.
- Ensures product quality by identifying defective items before they reach customers and maintaining manufacturing standards.

Where it is used

Healthcare: flags tumours in X-rays and MRI scans

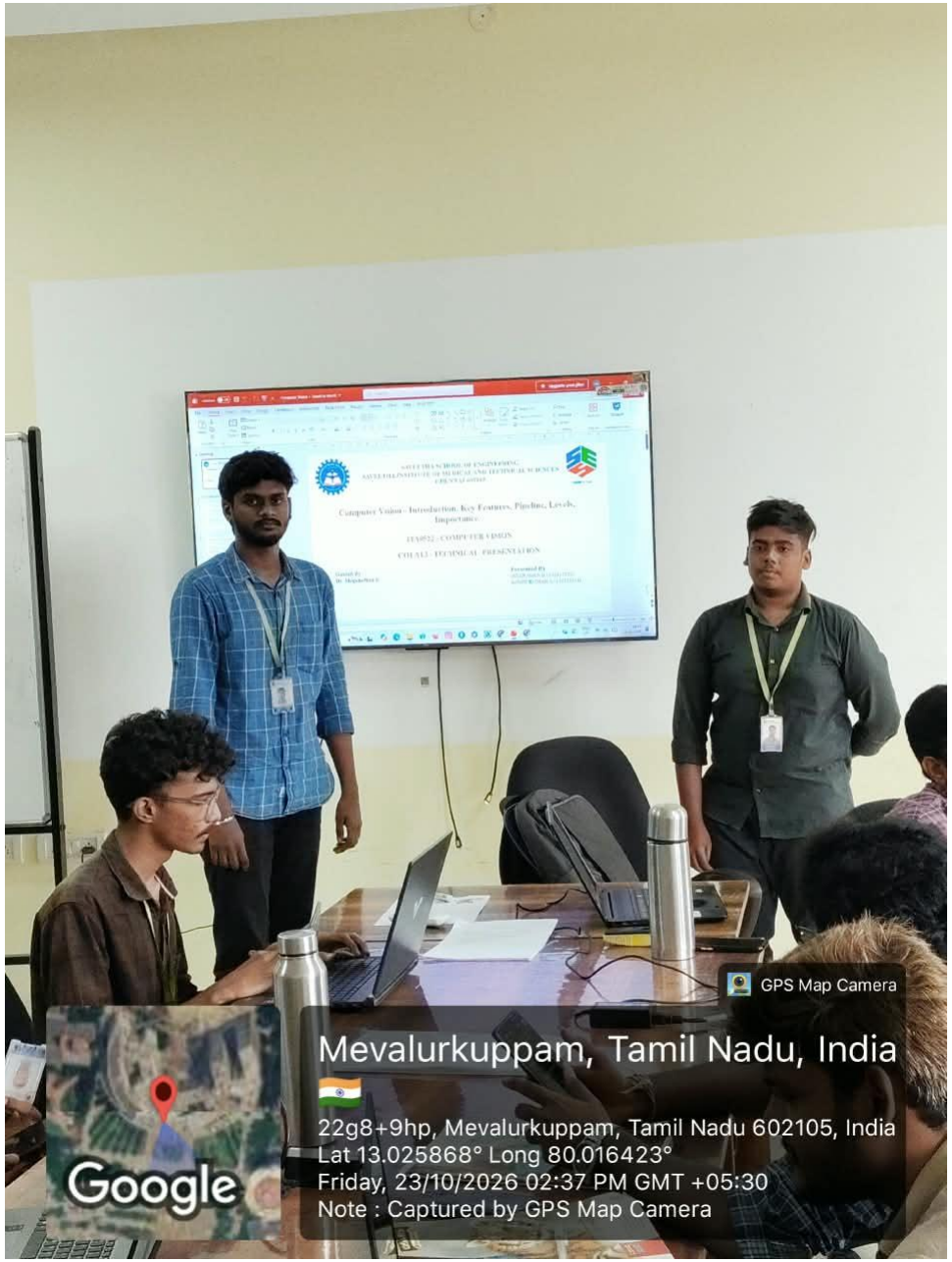
Autonomous driving: detects lanes, signs and pedestrians

Retail: cashierless checkout and shelf monitoring

Agriculture: spots crop disease from drone images

Security: face recognition and anomaly detection

THANK YOU


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